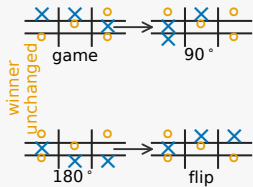
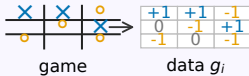


$$D_4 = \langle r, f \rangle$$



Symmetry



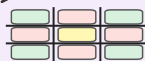
$$x_i = \frac{2\pi}{3} g_i = R_X(x_i)$$

x_0	x_1	x_2
x_7	x_8	x_3
x_6	x_5	x_4

Encoding

single-qubit gates

entangling gates



$H \leq D_4$: share by qubit and pair orbits

$$R_X(\theta_1)R_Y(\theta_2)$$

$$CRY(\theta)$$

winning-line triples

edge + line gates



$$ZZZ(\theta)$$

$$CCR_Z(\theta)$$

$$+ \text{ edge } CRY$$

Equivariant gateset